

A Technical Introduction to Reinforcement learning human feedback

Section 1

We finally train an optimal policy π^* that maximizes the objective function: $\pi^* = \arg \max_{\pi} \mathbb{E}_{(x,y) \sim D} [\pi^*(x,y) - \beta \log(\pi(y|x) \pi(\text{SFT}(y|x)))]$

Section 2

$$L_{\text{PPO-ptx}}(\phi) := \mathbb{E}_{(x,y) \sim D} [\phi^T \text{RL}[\min(\phi^T \text{RL}(y|x) \phi^T \text{RL}(y|x) A(x,y), \text{clip}(\phi^T \text{RL}(y|x) \phi^T \text{RL}(y|x), 1 - \epsilon, 1 + \epsilon) A(x,y)) - \beta \log(\pi(y|x) \pi(\text{SFT}(y|x)))] + \gamma \mathbb{E}_{x \sim D} [\log(\pi(\phi(x)))]$$

Section 3

Limitations RLHF suffers from challenges with collecting human feedback, learning a reward model, and optimizing the policy. Compared to data collection for techniques like unsupervised or self-supervised learning, collecting data for RLHF is less scalable and more expensive. Its quality and consistency may vary depending on the task, interface, and the preferences and biases of individual humans. The effectiveness of RLHF depends on the quality of human feedback. For instance, the model may become biased, favoring certain groups over others, if the feedback lacks impartiality, is inconsistent, or is incorrect. There is a risk of overfitting, where the model memorizes specific feedback examples instead of learning to generalize. For instance, feedback predominantly from a specific demographic might lead the model to learn peculiarities or noise, along with the intended alignment. Excessive alignment to the specific feedback it received (that is, to the bias therein) can lead to the model performing sub-optimally in new contexts or when used by different groups. A single reward function cannot always represent the opinions of diverse groups of people. Even with a representative sample, conflicting views and preferences may result in the reward model favoring the majority's opinion, potentially disadvantaging

underrepresented groups. In some cases, as is possible in regular reinforcement learning, there may be a risk of the model learning to manipulate the feedback process or game the system to achieve higher rewards rather than genuinely improving its performance. In the case of RLHF, a model may learn to exploit the fact that it is rewarded for what is evaluated positively and not necessarily for what is actually good, which can lead to it learning to persuade and manipulate. For example, models might learn that apparent confidence, even if inaccurate, garners higher rewards. Such behavior, if unchecked, is not just incentivized but can cause significant deployment issues due to the model's potential to mislead. Studies have found that humans are not skilled at identifying mistakes in LLM outputs in complex tasks; therefore, models learning to generate confident-sounding yet incorrect text can lead to significant issues when deployed.

Section 4

Applications RLHF has been applied to various domains of natural language processing (NLP), such as conversational agents, text summarization, and natural language understanding. Ordinary reinforcement learning, in which agents learn from their actions based on a predefined "reward function", is difficult to apply to NLP tasks because the rewards tend to be difficult to define or measure, especially when dealing with complex tasks that involve human values or preferences. RLHF can steer NLP models, in particular language models, to provide answers that align with human preferences with regard to such tasks by capturing their preferences beforehand in the reward model. This results in a model capable of generating more relevant responses and rejecting inappropriate or irrelevant queries. Some notable examples of RLHF-trained language models are OpenAI's ChatGPT (and its predecessor InstructGPT), DeepMind's Sparrow, Google's Gemini, and Anthropic's Claude. In computer vision, RLHF has also been used to align text-to-image models. Studies that successfully used RLHF for this goal have noted that the use of KL regularization in RLHF, which aims to prevent the learned policy from straying too far from the unaligned model, helped to stabilize the training process by reducing overfitting to the reward model. The final image outputs from models trained with KL regularization were noted to be of significantly higher quality than those trained without. Other methods tried to incorporate the feedback through more direct training—based on maximizing the reward without the use of reinforcement learning—but conceded that an RLHF-based approach would likely perform better due to the online sample generation used in RLHF during updates as well as the aforementioned KL regularization over the prior model, which mitigates overfitting to the reward function. RLHF was initially applied to other areas, such as the development of video game bots and tasks in simulated robotics. For example, OpenAI and DeepMind trained agents to play Atari games based on human

preferences. In classical RL-based training of such bots, the reward function is simply correlated to how well the agent is performing in the game, usually using metrics like the in-game score. In comparison, in RLHF, a human is periodically presented with two clips of the agent's behavior in the game and must decide which one looks better. This approach can teach agents to perform at a competitive level without ever having access to their score. In fact, it was shown that RLHF can sometimes lead to superior performance over RL with score metrics because the human's preferences can contain more useful information than performance-based metrics. The agents achieved strong performance in many of the environments tested, often surpassing human performance.